gGA Dimer Scans: U/t Dependence at $\Delta V = 0$ and $\Delta V = 2$

(Ghost Multiplicities $N_g = 2$ and $N_g = 4$, Direct-Minimization Solver)

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1 Total energy and error vs. exact: $N_g = 2$ and $N_g = 4$

Figures 1 and 2 show the variational energy E_{var}/t of the $N_g = 2$ and $N_g = 4$ solutions together with the exact (ED) energy, and the error $E_{\text{var}} - E_{\text{ED}}$ on a logarithmic scale, for $\Delta V = 0$ and $\Delta V = 2$ respectively. For $\Delta V = 2$ the plotted gGA energies are the cross-checked ("confirmed") solutions. The two N_g curves are indistinguishable on this scale. At $\Delta V = 0$, $\max |E(N_g{=}4) - E(N_g{=}2)| = 3 \times 10^{-8}$. At $\Delta V = 2$ the two agree to better than 10^{-6} for $U/t \lesssim 6.2$, while for $U/t \gtrsim 6.3$ the $N_g = 4$ energy is lower by up to 1.3×10^{-4} (at $U/t = 8.4$); this is still negligible against the error to the exact energy, ≈ 0.3–0.4 there (see Sec. 3).

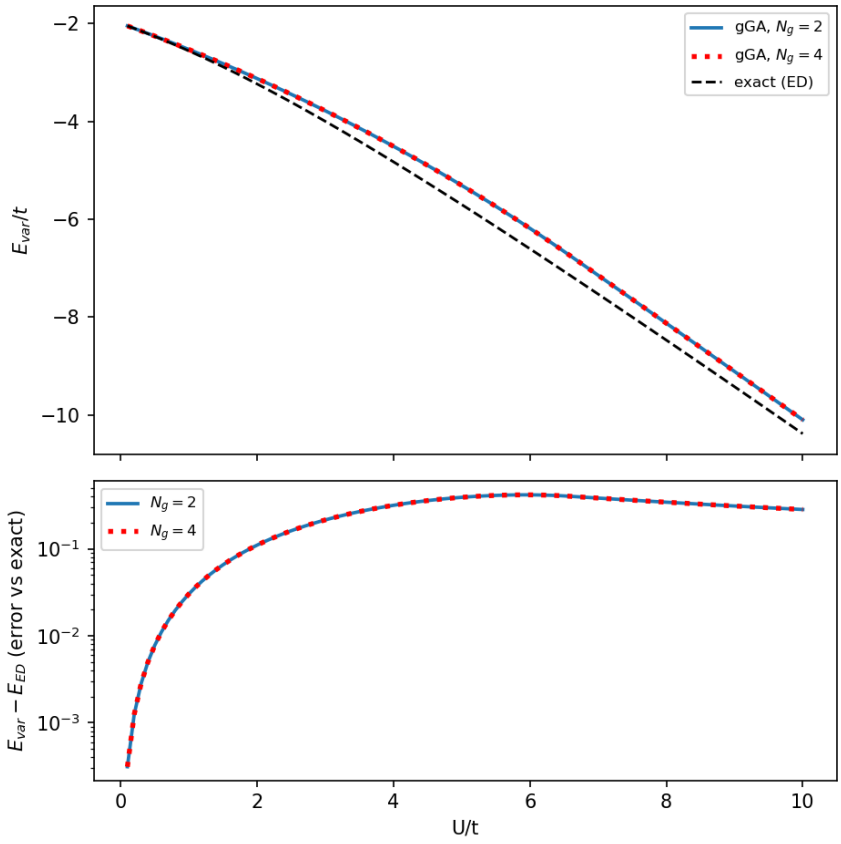


Figure 1: $\Delta V = 0$: E_{var}/t (top) and error $E_{\mathrm{var}} - E_{\mathrm{ED}}$ (bottom) vs. U/t for $N_g = 2$ and $N_g = 4$.

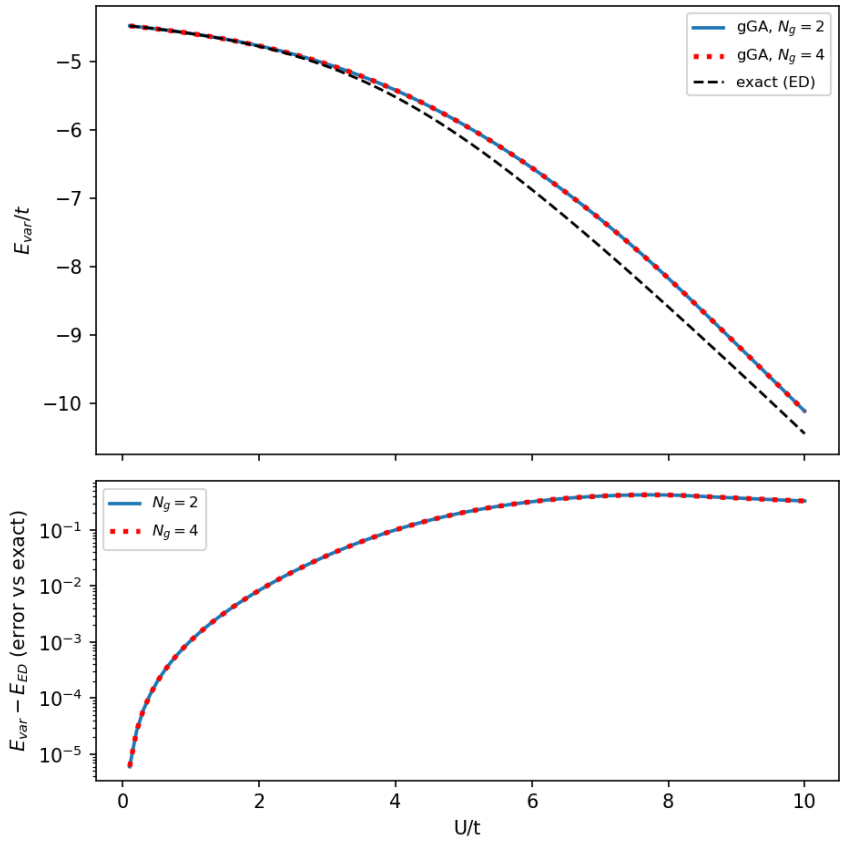
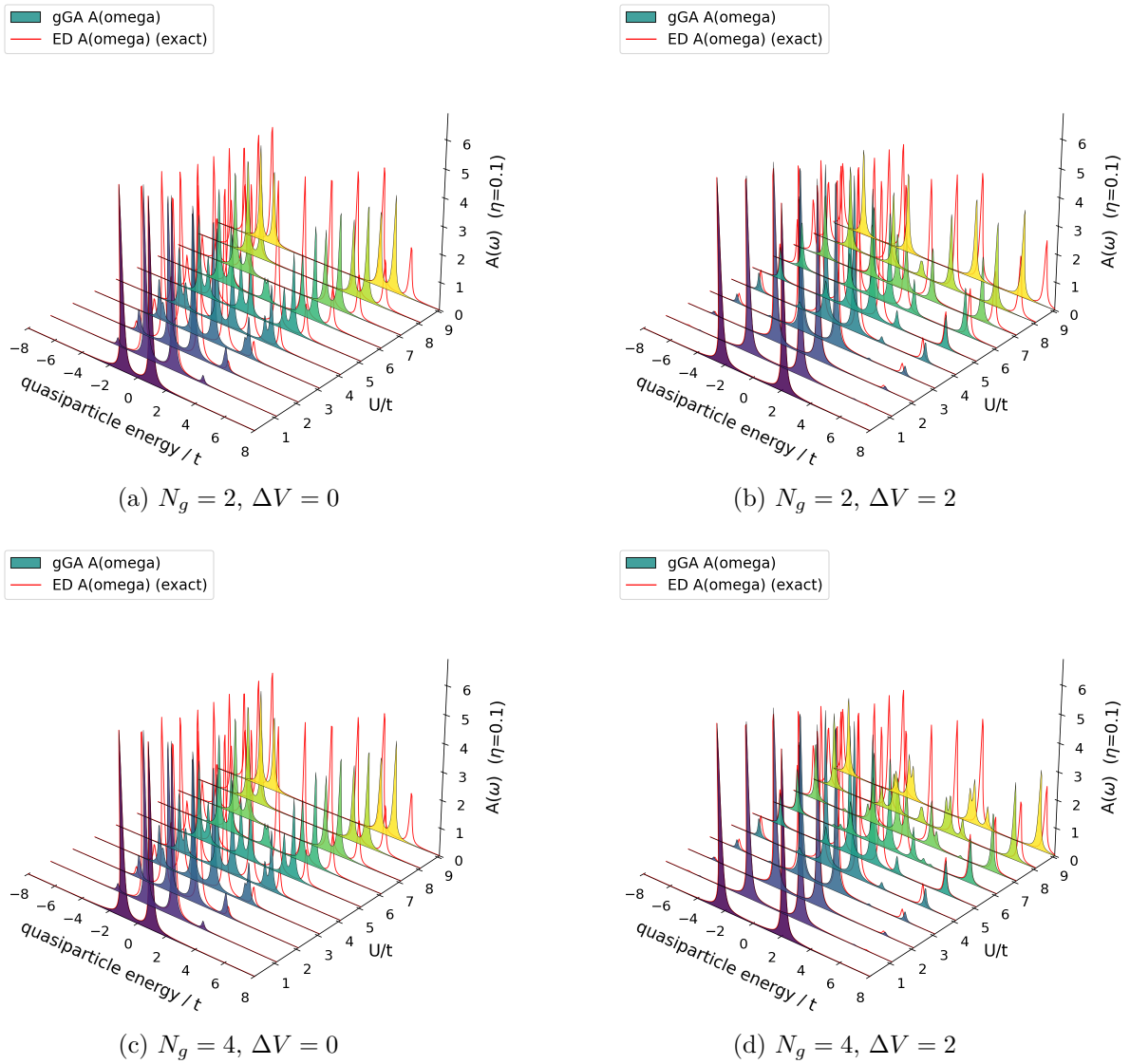


Figure 2: $\Delta V = 2$: E_{var}/t (top) and error $E_{\text{var}} - E_{\text{ED}}$ (bottom) vs. U/t for $N_g = 2$ and $N_g = 4$.

2 Spectra and spectral-function movies: $N_g = 2$ and $N_g = 4$

2.1 3D view

Figure 3 shows, for each $(N_g, \Delta V)$ combination, the gGA spectral function $A(\omega)$ (filled ridgelines, built from the quasiparticle poles and weights of the converged solution, broadening $\eta = 0.1$) as a function of U/t, together with the exact (ED) spectral function (red lines). Rows are $N_g = 2$ and $N_g = 4$, columns are $\Delta V = 0$ and $\Delta V = 2$. The $\Delta V = 2$ panels use the cross-checked solutions.



(a) $N_g = 2$, $\Delta V = 0$

(b) $N_g = 2$, $\Delta V = 2$

(c) $N_g = 4$, $\Delta V = 0$

(d) $N_g = 4$, $\Delta V = 2$

Figure 3: gGA spectral function $A(\omega)$ (filled) vs. ED (red) as a function of U/t for $N_g = 2$ (top) and $N_g = 4$ (bottom), at $\Delta V = 0$ (left) and $\Delta V = 2$ (right).

Analysis. The number of quasiparticle branches is $2(1 + N_g)$: six for $N_g = 2$ and ten for $N_g = 4$ (not a doubling). In all four panels the gGA peaks coincide with the ED ones at small U/t and move away from them as U/t grows, consistent with the error in the total energy growing with U/t (Sec. 1). The $N_g = 4$ ridgelines are visually indistinguishable from the $N_g = 2$ ones at the same ΔV: the extra branches carry no visible spectral weight, in line with the energies, so the gGA peaks do not follow the ED peaks more closely over the scanned range; the additional branches are the ones discussed as "sleeping" ghosts in Sec. 3. The effect of ΔV is visible in the same way for both N_g: at $\Delta V = 2$ the branches are well separated at small U/t and bunch together as U/t increases, with a rearrangement between $U/t = 0.6$ and 0.7 (Sec. 3).

2.2 Spectral function movies

- $N_g = 2$, $\Delta V = 0$: spectral-function movie

- $N_g = 2$, $\Delta V = 2$: spectral-function movie
- $N_g = 4$, $\Delta V = 0$: spectral-function movie
- $N_g = 4$, $\Delta V = 2$: spectral-function movie

(Links open the movie with the PDF viewer's associated application, if the viewer supports `run:` links.)

3 Discussion: $\Delta V = 0$ vs. $\Delta V = 2$

Weak coupling: ΔV dominates. At small U/t the two scans differ substantially. At $U/t = 0.1$, $E_{\mathrm{var}}/t = -2.0503$ for $\Delta V = 0$ but -4.4823 for $\Delta V = 2$ (both N_g). This is the expected $U \to 0$ limit: with $\Delta V = 0$ the dimer's noninteracting energy is $-2t = -2$ plus a small $O(U)$ correction, while at $\Delta V = 2$ the site-splitting opens a one-body level gap and the $U \to 0$ energy approaches the closed form $E = -2\sqrt{t^2 + \Delta V^2} = -2\sqrt{5} \approx -4.472$. The gap between the two curves is largest here, $\Delta E \approx 2.43$, because ΔV acts as an unscreened one-body potential that the (still weak) correlations barely renormalize.

Strong coupling: the curves converge. As U/t grows the two energies draw close together: at $U/t = 10$, $E_{\mathrm{var}}/t = -10.101$ ($\Delta V = 0$) vs. -10.120 ($\Delta V = 2$), a difference of only ≈ 0.02 – two orders of magnitude smaller than at weak coupling. In the Mott-like regime the total energy is governed almost entirely by U (both sites approach single occupancy), and the fixed $\Delta V = 2$ site-splitting becomes a small perturbation on top of the dominant Hubbard physics. The quasiparticle spectra (3D plots, Fig. 3) show the same trend: the $\Delta V = 2$ branches, well separated at small U, bunch together with the $\Delta V = 0$-like structure as U/t increases.

Level crossing at $\Delta V = 2$. The $\Delta V = 2$ quasiparticle energies show a sharp rearrangement between $U/t = 0.6$ and $U/t = 0.7$ (e.g. $N_g = 2$: $\epsilon^{qp} \approx -2.00 \to -3.12$), visible as a kink in the 3D plots; this is the level crossing where the ΔV-induced one-body gap stops being the dominant scale and the Hubbard repulsion takes over.

Ghost multiplicity ($N_g = 2$ vs. $N_g = 4$). At both $\Delta V = 0$ and $\Delta V = 2$, doubling the ghost multiplicity from $N_g = 2$ to $N_g = 4$ leaves the lowest variational energy numerically unchanged at $\Delta V = 0$ over the whole scanned range, and at $\Delta V = 2$ for $U/t \lesssim 6.2$ (for larger U/t the $N_g = 4$ energy is lower by at most 1.3×10^{-4}, Sec. 1): at $\Delta V = 0$, $E(N_g{=}4) - E(N_g{=}2) = 0$ to the printed precision at essentially every U/t (largest deviation $\sim 3 \times 10^{-8}$, at the level of solver tolerance), and the confirmed energies show the same agreement at $\Delta V = 2$ (e.g. $U/t = 1.0$: -4.593786 vs. -4.593786). The extra ghost degrees of freedom introduced at $N_g = 4$ therefore stay essentially energetically inert for this dimer at both values of ΔV in this U range – consistent with the known $N_g \geq 2$ "sleeping ghost" caveat: the additional branches exist in the quasiparticle spectrum (compare Fig. 3, which shows ten instead of six ϵ_n^{qp} branches at $N_g = 4$) but do not lower the ground-state energy below what $N_g = 2$ already finds.